Basic Linear Regression

Line of best fit go brrr

Your Name

October 28, 2026

The Ultimate Goal

Fit a line to 2D Data

- Problem Introduction

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- Initial Quantification of Problem

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- (After) Practical Workshop

The Points

Definition

Dataset of points $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N} = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(N)}, y^{(N)})\}$

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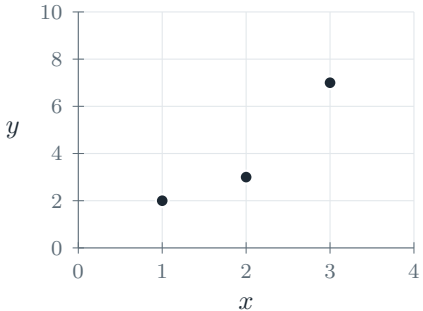
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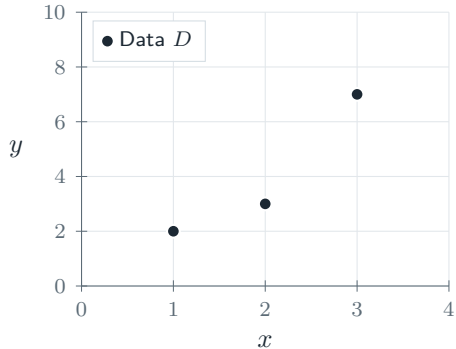
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A line as a function of x, with m and c given:

$$f(x \mid m, c) = mx + c$$

Think: I need m and c before I can compute $f(x)$.



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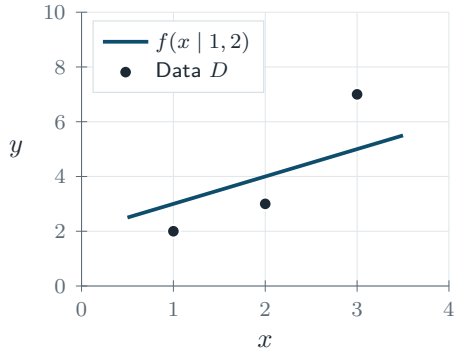
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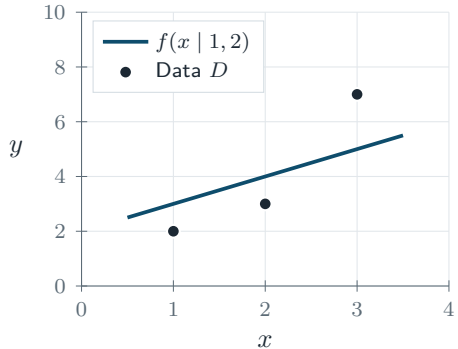
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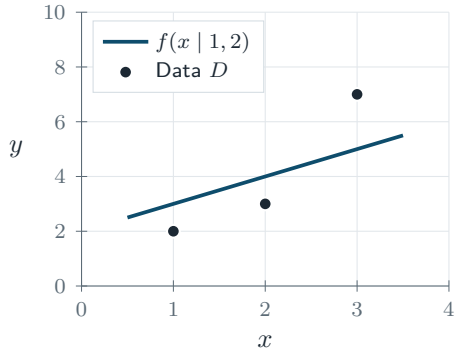


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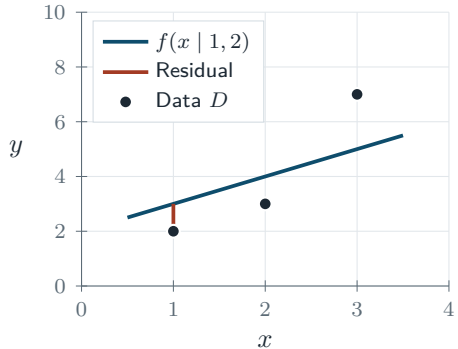


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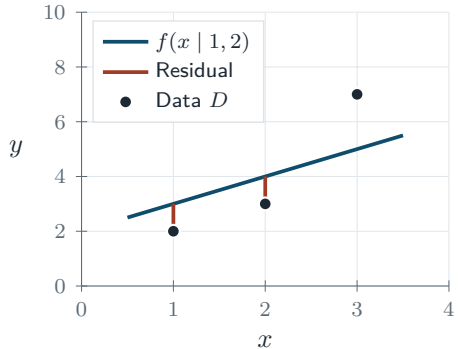


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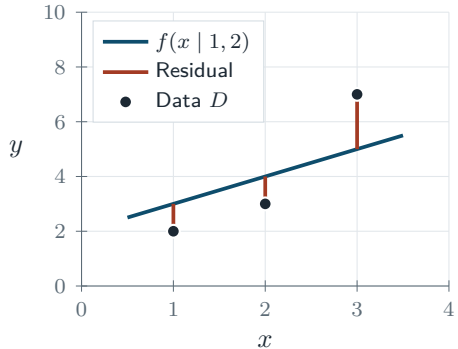


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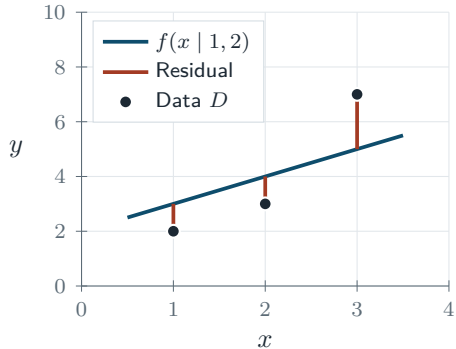
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A good line makes all residuals small.

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Choose m and c to minimise:

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10 Minute Calculus Pulse Check

Do not simplify your answers.

Problem 1

$$g(x) = (3x + 1)^{21} + (x^5 + 6x)^{15}$$

Find $\dfrac{\mathrm{d}g}{\mathrm{d}x}$.

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Problem 2

$$h(m) = \sum_{i=1}^{500} (2m^7 - 5m^3)^i$$

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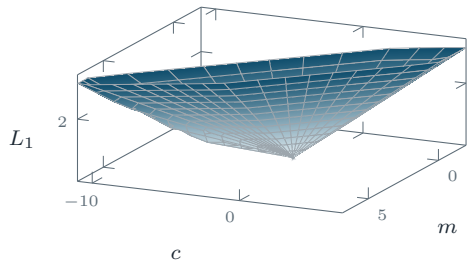
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$$= \sum_{i=1}^{500} i(2m^7 - 5m^3)^{i-1}(14m^6 - 15m^2) \qquad \text{chain} \qquad (2)$$

The Partial Derivative

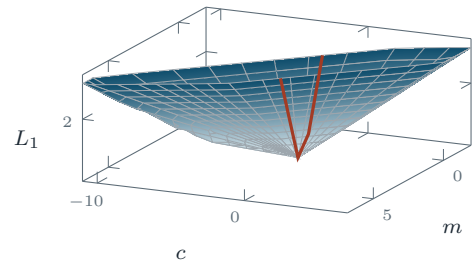
For multivariable functions like $L_1(m, c)$.

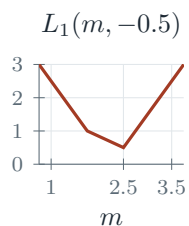


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$\frac{\partial L_1}{\partial m}$ Take c constant, differentiate w.r.t m.



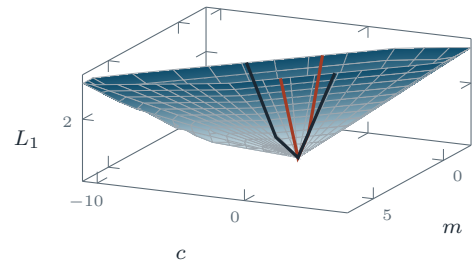


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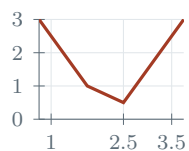
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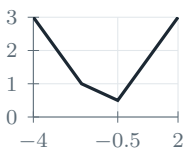
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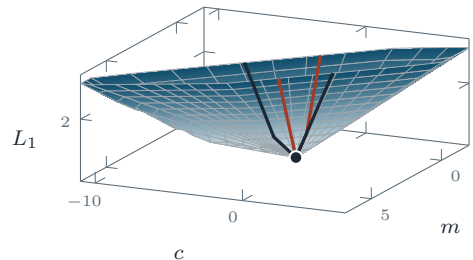
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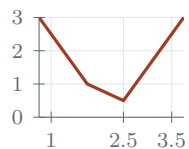
$\frac{\partial L_1}{\partial c}$ Take m constant, differentiate w.r.t c.

The minimum L_1 is achieved through a choice of m and c such that

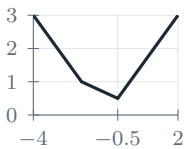
$$\frac{\partial L_1}{\partial m} = \frac{\partial L_1}{\partial c} = 0$$



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Calculating $\frac{\partial L_1}{\partial m}$

$$\frac{\partial L_1}{\partial m} = \frac{1}{N} \cdot \frac{\partial}{\partial m} \sum_{i=1}^{N} |r(i \mid m, c)|$$

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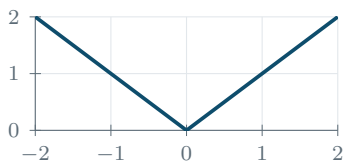
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$$f(t) = |t|$$

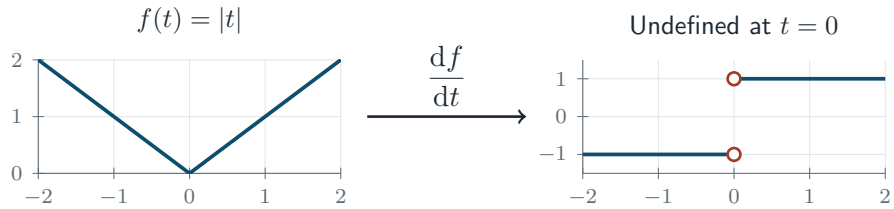


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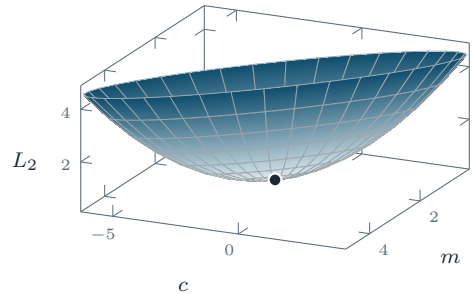
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$$L_2(m, c) = \frac{1}{N} \sum_{i=1}^{N} r(i \mid m, c)^2 = \frac{1}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big)^2$$

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L_2 over D: a smooth bowl with one lowest point.

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&= \frac{2}{N}\sum_{i=1}^{N}\Big(f(x^{(i)} \mid m, c) - y^{(i)}\Big)x^{(i)}
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$$\frac{\partial L_2}{\partial m} = \frac{2}{N}\sum_{i=1}^{N}\Big(f(x^{(i)} \mid m, c) - y^{(i)}\Big)x^{(i)}$$

As an exercise, you can show that

$$\frac{\partial L_2}{\partial c} = \frac{2}{N}\sum_{i=1}^{N}\Big(f(x^{(i)} \mid m, c) - y^{(i)}\Big)$$

Finding m and c

Set both partial derivatives to 0.

$$0 = \frac{2}{N} \sum_{i=1}^{N} \Big(f(x^{(i)} \mid m, c) - y^{(i)} \Big) x^{(i)}$$

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A lot of re-arranging occurs, which takes too long. Hence left as exercise.

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End result:

$$m = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\overline{x^2} - \overline{x}^2}$$

$$c = \overline{y} - m\overline{x}$$

Finding m and c

End result:

$$m = \frac{\overline{xy} - \overline{x} \cdot \overline{y}}{\overline{x^2} - \overline{x}^2}$$

$$c = \overline{y} - m\overline{x}$$

Where:

$$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x^{(i)} \qquad \overline{y} = \frac{1}{N}\sum_{i=1}^{N} y^{(i)}$$

$$\overline{xy} = \frac{1}{N}\sum_{i=1}^{N} x^{(i)}y^{(i)} \qquad \overline{x^2} = \frac{1}{N}\sum_{i=1}^{N} \left(x^{(i)}\right)^2$$

Application to our dataset

i	$x^{(i)}$	$y^{(i)}$
1	1	2
2	2	3
3	3	7

$\overline{x}$

$\overline{y}$

$\overline{xy}$

$\overline{x^2}$

Application to our dataset

i	$x^{(i)}$	$y^{(i)}$
1	1	2
2	2	3
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$$\overline{x} = \frac{1+2+3}{3} = 2$$

$$\overline{y} = \frac{2+3+7}{3} = 4$$

$\overline{xy}$

$\overline{x^2}$

Application to our dataset

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$$\overline{x} = \frac{1+2+3}{3} = 2 \qquad \overline{y} = \frac{2+3+7}{3} = 4$$

$$\overline{xy} = \frac{1 \cdot 2 + 2 \cdot 3 + 3 \cdot 7}{3} = \frac{29}{3} \qquad \overline{x^2} = \frac{1^2 + 2^2 + 3^2}{3} = \frac{14}{3}$$

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$$m = \frac{\overline{xy} - \overline{x}\cdot\overline{y}}{\overline{x^2} - \overline{x}^2} = \frac{29/3 - 2\cdot 4}{14/3 - 2^2} = \frac{5/3}{2/3} = 2.5$$

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$$c = \overline{y} - m\overline{x} = 4 - 2.5\cdot 2 = -1$$

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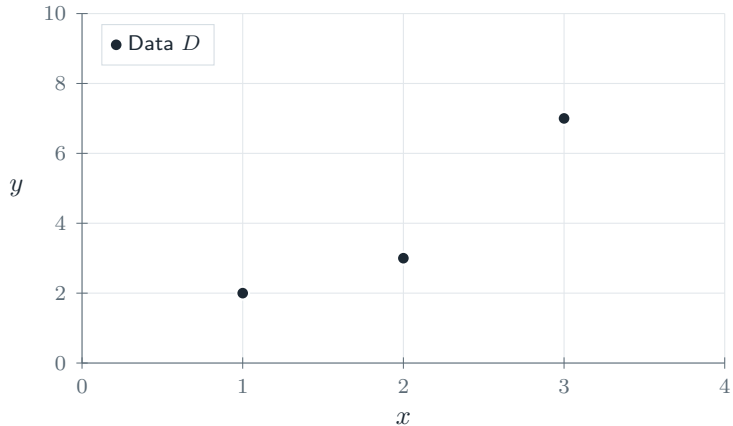
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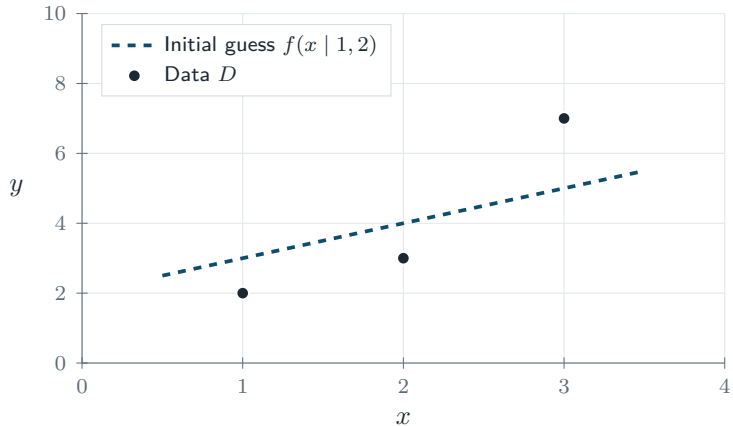
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Line of best fit: $f(x \mid 2.5, -1) = 2.5x - 1$

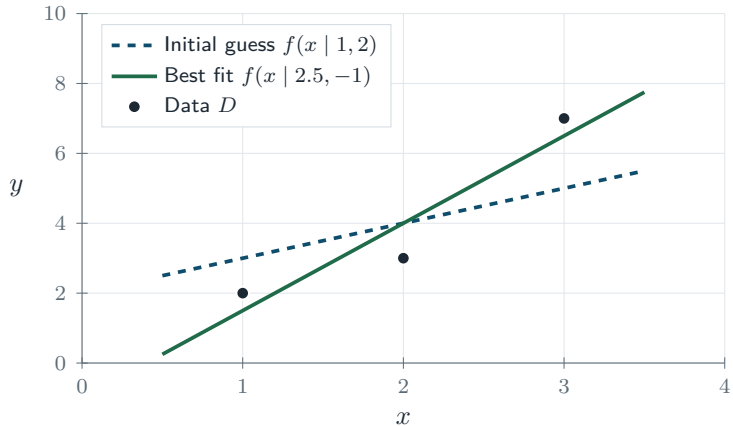
Visualisation



Visualisation



Visualisation



Comparing L_2

Initial guess $f(x \mid 1, 2)$

i	$x^{(i)}$	$y^{(i)}$	$r(i \mid 1, 2)$	r^2
1	1	2	$3 - 2 = 1$	1
2	2	3	$4 - 3 = 1$	1
3	3	7	$5 - 7 = -2$	4

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$$L_2(1, 2) = \frac{1 + 1 + 4}{3} = 2$$

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$$L_2(1, 2) = \frac{1 + 1 + 4}{3} = 2$$

Best fit $f(x \mid 2.5, -1)$

i	$x^{(i)}$	$y^{(i)}$	$r(i \mid 2.5, -1)$	r^2
1	1	2	$1.5 - 2 = -0.5$	0.25
2	2	3	$4 - 3 = 1$	1
3	3	7	$6.5 - 7 = -0.5$	0.25

Comparing L_2

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$0.5 < 2$, so the best fit beats the initial guess.

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$0.5 < 2$, so the best fit beats the initial guess.

L_2 is a bowl - the point where both partial derivatives are 0 is its lowest point: no other m and c give smaller L_2.